

APEX Device Class — Wayfinding

APEX — Adaptive Payload EXchange

Status: Draft | Version: 0.6.0 | Scope: Wayfinding device class (`traffic_type = 3`)

1. Overview

The Wayfinding class covers Devices that produce **navigational cues** for the operator — *"point me in this direction"* updates the Host renders on its display, typically as a labeled arrow per target. The Host shows the direction and the description; the operator decides how to act.

The protocol is **agnostic to the source of the cue**. A wayfinding Device may compute its targets from anything it likes; only the resulting bearing and label appear on the wire. Each update is a small set of measurements about a **target**: a bearing the operator should look toward, optionally an elevation and a proximity value, and a short human-readable descriptor (e.g. *"WAYPOINT#1"*, *"Target #1"*, *"01"*).

This class supports:

- Multiple simultaneous targets, each addressed by a small index (1..N).
- A short ASCII descriptor per target to give the operator context.
- Optional per-measurement fields (elevation, proximity) that a Device may include or omit; the Host parses only what the Device declared.
- Stream-and-forget operation — the Host does not poll, and there is no per-update acknowledgement.

This class does not support (out of scope for the current version; future class candidates):

- Bidirectional negotiation about *which* target the operator is following.
- Bearing values at sub-degree precision. Bearings are reported in whole degrees (§3.1); a future version may add a higher-precision mode.

This document covers **both sides** of the class — Device role and Host role. Discovery, framing, and capability exchange at the bus level are out of scope and are handled by Core. All frames in this class carry `traffic_type = 3` in the APEX outer header.

2. Roles

2.1. Device

The Device declares the shape of the updates it will produce — the number of target slots it uses, which optional measurement fields it includes, and the maximum descriptor length — via the CAPABILITY frame (§4.3). On ACCEPTED (§4.5), the Device streams TARGET_UPDATE frames (§4.6) at its own cadence. The Device decides when to send; the Host does not poll. A Device may add, refresh, or clear targets at any time while in ACTIVE.

2.2. Host

The Host receives CAPABILITY, replies with CONFIG (§4.4) (acceptance is essentially unconditional in the current version; the CONFIG frame exists to surface a clear malformed-capability case), and consumes incoming TARGET_UPDATE frames. The Host presents the cues however the platform sees fit — typically an OSD arrow per target index with the descriptor as a label.

The Host MUST NOT send TARGET_UPDATE frames. Wayfinding is one-way Device → Host.

3. Bearing Reference and Measurement Fields

3.1. Bearing reference

Bearings are reported **relative to the airframe's current heading**, in whole degrees, with 0° straight ahead and increasing clockwise. A bearing of 90° points to the operator's right, 180° aft, 270° to the left.

Value	Meaning
0 – 359	Valid bearing in degrees relative to heading.
0xFFFF	Clear — remove the target at the given index from the Host's display (§4.6).
360 – 0xFFFE	Reserved. A Device MUST NOT send these; a Host receiving one MUST treat the TARGET_UPDATE as malformed and drop it silently (APEX — Core §3.8).

A Device that knows its targets in an absolute frame is responsible for converting to heading-relative before transmitting. The Host is not required to provide the heading.

3.2. Optional measurement fields

A Device may include any of the following additional measurements in each TARGET_UPDATE. Which ones are included is declared **once** in the CAPABILITY frame (§4.3) as a bitmask, and every TARGET_UPDATE from that Device carries exactly the declared set, in the order below.

Bit	Field	Width	Description
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0	ELEVATION	i16	Elevation above horizontal, in whole degrees ($-90 \dots +90$). Reserved bit pattern <code>0x8000</code> means <i>unknown</i> .
1	PROXIMITY	i16	A measure of how close the target is, in payload-defined units. The protocol does not assign units or interpret the value — the Device declares the unit and polarity (i.e. whether higher or lower means closer) once in the CAPABILITY frame (§4.3). Reserved bit pattern <code>0x8000</code> means <i>unknown</i> .
2 – 7	(reserved)	—	MUST be 0.

Bits 2–7 are reserved and a Device MUST NOT set them in the current version; a Host that receives a CAPABILITY with any reserved bit set replies `REJECT_MALFORMED` (§4.5).

A Device that includes no optional fields sets `optional_fields = 0`. A Device that includes elevation and proximity sets `optional_fields = 0b00000011`.

4. Message Format

The class follows the APEX framing model (APEX — Core §3): every frame carries `traffic_type = 3`, is COBS-framed, and uses the outer header from APEX — Core §3.1.1, with the inner payload carrying class-specific content. All multi-byte fields are little-endian (APEX — Core §3.1).

4.1. Inner payload sub-header

<code>class_msg_id</code>	Name	Direction	Meaning
0	(reserved)	—	Reserved; never sent.
1	CAPABILITY	Device → Host	Declares update shape (§4.3).
2	CONFIG	Host → Device	Accepts or implicitly rejects the capability (§4.4).
3	ACK	Device → Host	Acknowledges CONFIG (§4.5).
4	TARGET_UPDATE	Device → Host	One target's current measurements (§4.6).

4.2. Lifecycle overview

1. Core discovery completes for `device_class_req = 3` (APEX — Core §3.3). Class traffic on `traffic_type = 3` becomes valid.

2. Device immediately emits **CAPABILITY** (§4.3) — unprompted, exactly once.
3. Host emits **CONFIG** (§4.4) accepting the declared shape.
4. Device replies **ACK** (§4.5). On `ACCEPTED`, the session enters **ACTIVE**.
5. Device streams **TARGET_UPDATE** frames (§4.6) at its own cadence for the remainder of the session.

4.3. CAPABILITY frame (Device → Host)

Offset	Field	Width	Description
0	<code>class_msg_id</code>	u8	1 (CAPABILITY).
1	<code>class_spec_version</code>	u8	Wayfinding-class spec revision the Device implements. 0 = the revision defined by this document.
2	<code>max_targets</code>	u8	Number of simultaneously addressable target slots. 1..16. The Host uses indices 1..max_targets in TARGET_UPDATE frames.
3	<code>optional_fields</code>	u8	Bitmask of optional measurement fields the Device includes in every TARGET_UPDATE (§3.2).
4	<code>descriptor_max_bytes</code>	u8	Maximum length, in bytes, of the descriptor field the Device will ever emit. 0..32. A value of 0 means the Device does not emit descriptors.
5	<code>proximity_polarity</code>	u8	Tells the Host which direction of change means <i>the target is getting closer</i> . 0 = LOWER_IS_CLOSER (e.g. meters, range — the value decreases as the operator approaches). 1 = HIGHER_IS_CLOSER (e.g. RSSI, signal strength — the value increases as the operator approaches). Present only when bit 1 of optional_fields is set (§3.2); omitted otherwise.
6..13	<code>proximity_unit</code>	char[8]	ASCII unit label for the PROXIMITY field, NUL-padded to 8 bytes. The Host displays this string verbatim alongside the value (typical labels: "m", "km", "dBm", "RSSI"). Present only when bit 1 of optional_fields is set; omitted otherwise. The string is the last field in CAPABILITY so a future revision can promote it to a variable-length representation without disturbing earlier offsets.

A `max_targets` of 0 or `max_targets > 16` is malformed; the Host MUST reply `REJECT_MALFORMED` (§4.5). A `descriptor_max_bytes > 32` is malformed for the same reason. A `proximity_polarity` value other than 0 or 1 is malformed.

A `class_spec_version` value the Host does not understand should be treated as 0 — the Host parses the fields it knows. Future revisions of this spec will extend CAPABILITY only by appending fields, never by repurposing existing ones.

4.4. CONFIG frame (Host → Device)

Offset	Field	Width	Description
0	<code>class_msg_id</code>	u8	2 (CONFIG).

The CONFIG frame carries no body in the current version. Its purpose is to give the Host an explicit handshake step on which it may decide to reject the capability — see §4.5. A Host that accepts the capability sends CONFIG; a Host that finds it malformed sends nothing and lets the Device timeout to FAULT, or sends CONFIG and lets the Device's own validation produce `REJECT_MALFORMED`.

A future revision may add negotiation fields here (e.g. cadence caps, descriptor encoding selection).

4.5. ACK frame (Device → Host)

Offset	Field	Width	Description
0	<code>class_msg_id</code>	u8	3 (ACK).
1	<code>result</code>	u8	Result code (see below).

result values:

Value	Name	Meaning
0x00	ACCEPTED	The Device has entered ACTIVE; TARGET_UPDATE streaming begins.
0x01	REJECT_MALFORMED	The Device's own CAPABILITY was malformed (reserved bits set, <code>max_targets</code> out of range, etc.) or the Host's CONFIG was malformed.

A rejecting ACK is **terminal in the current version**: the Device transitions to FAULT (§4.7) and the operator must intervene. A Host that detects a malformed CAPABILITY SHOULD surface the reject to the operator; that is the only signal a passive operator has.

4.6. TARGET_UPDATE frame (Device → Host)

Offset	Field	Width	Description
0	<code>class_msg_id</code>	u8	4 (TARGET_UPDATE).

1	target_index	u8	Index of the target this update applies to. 1..max_targets .
2	bearing	u16	Heading-relative bearing in degrees (§3.1). 0..359 is a valid bearing; 0xFFFF clears the target.
4...	optional fields	variable	The optional measurement fields the Device declared in CAPABILITY (§3.2), in the bit order shown there: ELEVATION, then PROXIMITY. Only the declared fields are present.
(after)	descriptor_len	u8	Number of bytes of descriptor that follow. 0..descriptor_max_bytes . Omitted entirely if descriptor_max_bytes == 0 in CAPABILITY.
(after)	descriptor	descriptor_len × u8	ASCII descriptor for the target (no NUL terminator). May change between updates for the same target_index .

A TARGET_UPDATE with target_index == 0 or target_index > max_targets is malformed; the Host MUST drop it silently. A TARGET_UPDATE with descriptor_len > descriptor_max_bytes is malformed; the Host MUST drop it silently.

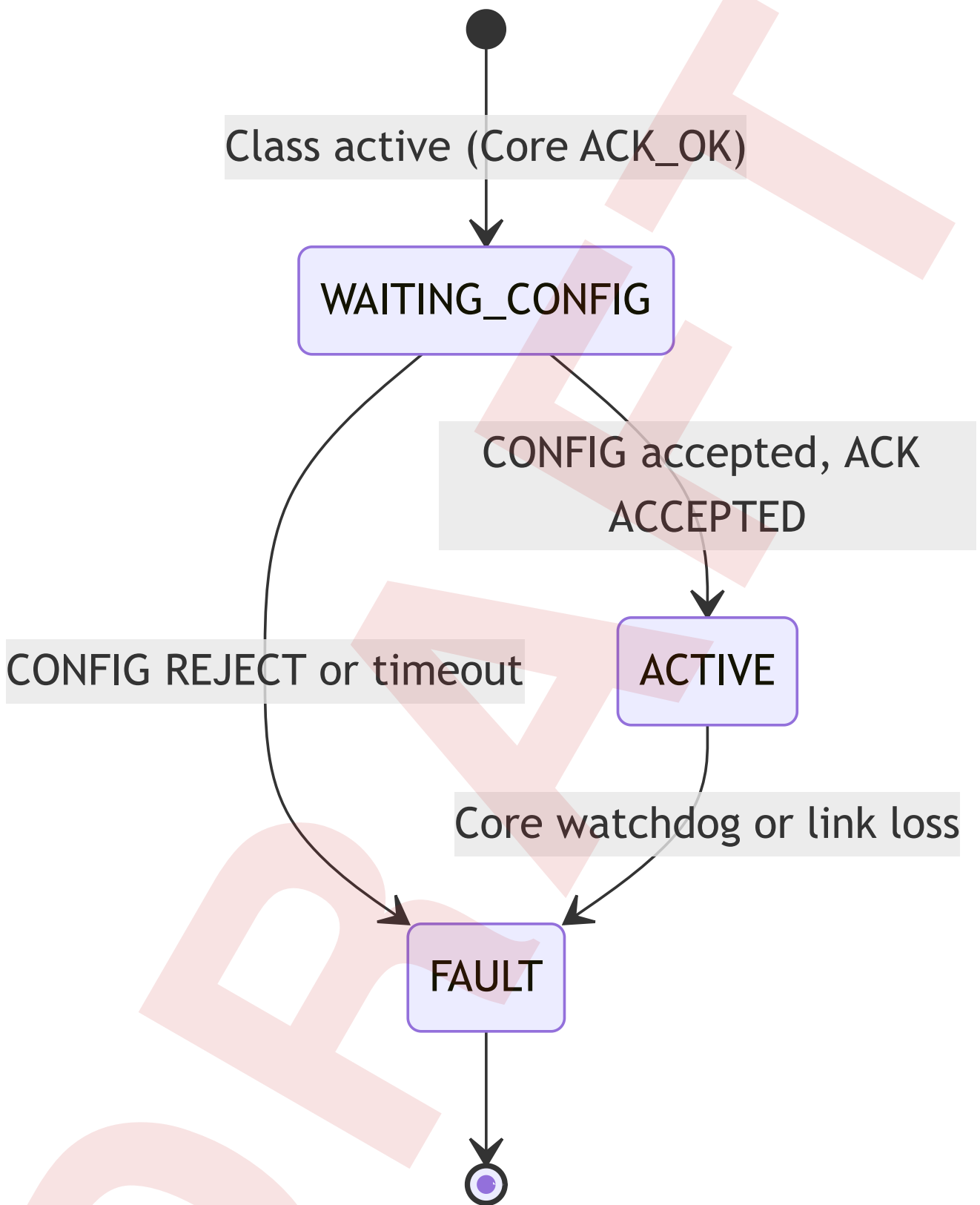
Clearing a target. A Device clears target *i* by sending a TARGET_UPDATE with target_index = *i* and bearing = 0xFFFF . The optional fields and descriptor SHOULD be omitted (i.e. the frame ends after bearing) when clearing, but the Host MUST tolerate a clear frame whose declared optional fields and descriptor are present and ignore their values.

Refreshing a target. The Device sends TARGET_UPDATE for the same target_index as often as it has new data; each update replaces the previous one for that index.

4.7. State machine (Device)

The Device runs the following minimal state machine.

Value	Name	Description
0x01	WAITING_CONFIG	CAPABILITY has been sent; awaiting CONFIG.
0x02	ACTIVE	CONFIG accepted; TARGET_UPDATE streaming.
0xFF	FAULT	CONFIG rejected, malformed traffic, or core-layer fault. Terminal in the current version.



The Host does not run an explicit state machine for this class beyond *pre-CONFIG* / *ACTIVE* per device — the device's Core lifecycle status ([APEX — Core §4](#)) tells it everything else it needs.

4.8. Reporting cadence and timing

- **CAPABILITY.** Emitted within **100 ms** of the class becoming active.

- **CONFIG.** The Host SHOULD emit CONFIG within **500 ms** of receiving CAPABILITY. A Device that has not received CONFIG within **5 s** transitions to FAULT.
- **ACK.** The Device MUST emit ACK within **200 ms** of receiving CONFIG.
- **TARGET_UPDATE.** Streaming begins immediately after the Device sends `ACCEPTED`. Cadence is payload-defined — typical wayfinding payloads emit updates at 5–25 Hz per target. The Device's TARGET_UPDATE traffic satisfies the [APEX — Core §3.5](#) 1 Hz transmit floor; the Host's HOST_STATE broadcast satisfies the same on the Host side. A Device with currently no live targets MUST still send at least one frame per second — the implicit Core heartbeat is sufficient.

5. Limitations

5.1. Bearing resolution

Bearings are whole degrees. This matches the on-display resolution of typical arrow renderers and is plenty for *point me there* cues. A payload whose internal bearing is more precise rounds before sending. A future revision of this class may add a higher-precision mode if real use cases demand it; the current version does not.

5.2. Target count

`max_targets` is capped at 16. Real wayfinding displays render only a handful of targets legibly; sixteen is a generous ceiling that bounds the worst-case CAPABILITY and TARGET_UPDATE frame size.

5.3. No Host → Device target feedback

The Host cannot tell the Device which target the operator is currently following, nor ask the Device to refresh a specific target. The current version is strictly one-way (Device → Host) after the CONFIG handshake. A future revision may add a feedback channel for use cases where the payload could benefit from operator attention.

6. Worked Example

A wayfinding payload that streams up to **3 targets** with a **proximity** measurement (reported in meters, lower-is-closer) and descriptors up to **7 bytes** long.

6.1. The example Device

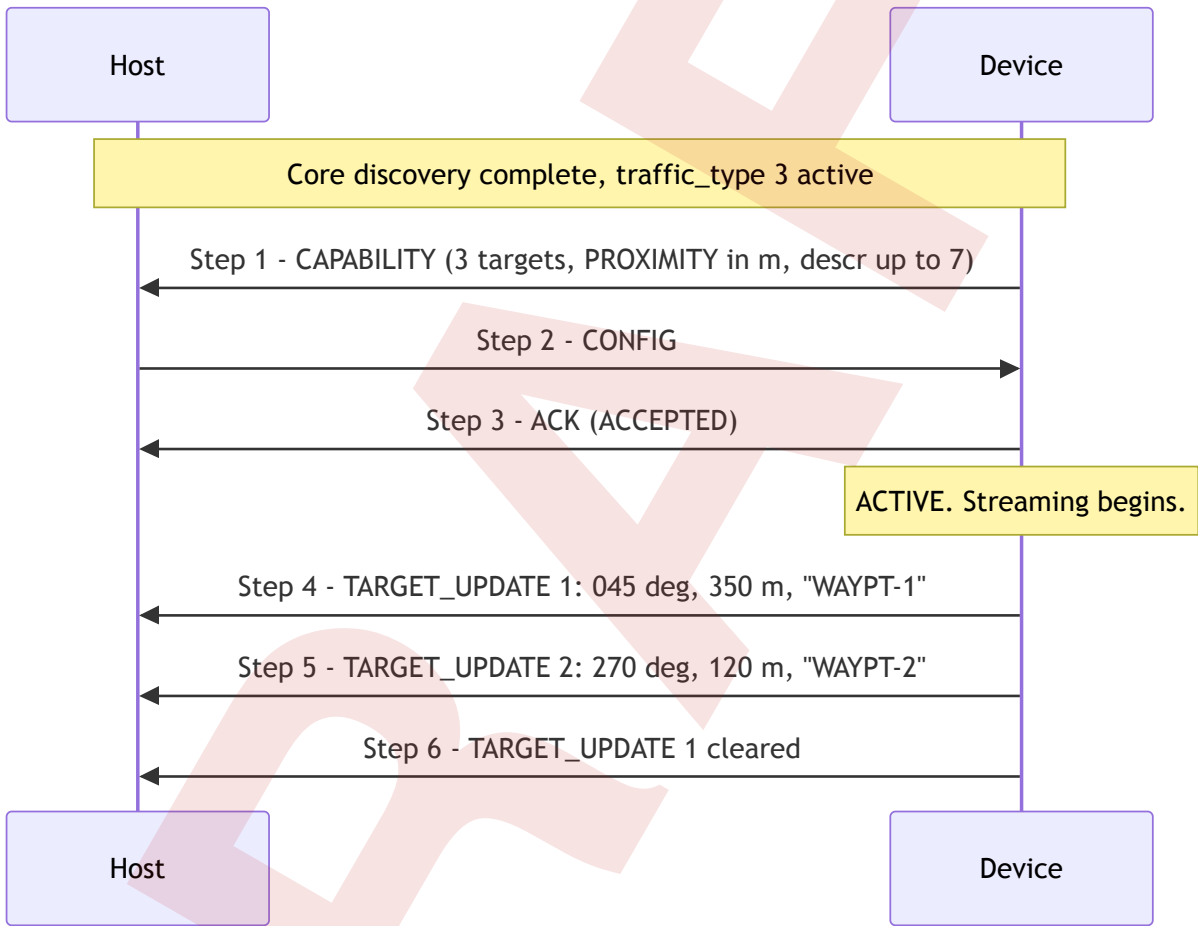
- `max_targets = 3`
- `optional_fields = 0b00000010` (PROXIMITY only)
- `descriptor_max_bytes = 7`

- `proximity_polarity = 0` (LOWER_IS_CLOSER)
- `proximity_unit = "m"`
- Has completed Core discovery and been assigned `device_id = 0x01`.

6.2. Frame notation

Frames below use the common notation defined in Core §3.1.5: each is shown as the bytes before CRC and COBS. `PV = 00` , `TT = 03` , `ID = 01` throughout.

6.3. Sequence



► Step 1 — Device emits CAPABILITY

Inner payload 14 bytes (`LN = 0E`).

00 03 01 0E 01 00 03 02 07 00 6D 00 00 00 00 00 00 00			
Byte(s)	Hex	Field	Value
0–3	00 03 01 0E	outer header	LN=0E (14)
4	01	class_msg_id	1 (CAPABILITY)

5	00	class_spec_version	0
6	03	max_targets	3
7	02	optional_fields	0b00000010 (PROXIMITY)
8	07	descriptor_max_bytes	7
9	00	proximity_polarity	0 (LOWER_IS_CLOSER)
10–17	6D 00 00 00 00 00 00 00	proximity_unit	"m" (NUL-padded to 8 bytes)

► Step 2 — Host emits CONFIG

Inner payload 1 byte (LN = 01).

```
00 03 01 01 02
```

Byte(s)	Hex	Field	Value
4	02	class_msg_id	2 (CONFIG)

► Step 3 — Device acknowledges

Inner payload 2 bytes (LN = 02).

```
00 03 01 02 03 00
```

Byte(s)	Hex	Field	Value
4	03	class_msg_id	3 (ACK)
5	00	result	ACCEPTED

The Device transitions WAITING_CONFIG → ACTIVE and begins streaming.

► Step 4 — TARGET_UPDATE for target 1: WAYPT-1 at 045°, 350 m

Inner payload 14 bytes (LN = 0E).

```
00 03 01 0E 04 01 2D 00 5E 01 07 57 41 59 50 54 2D 31
```

Byte(s)	Hex	Field	Value
4	04	class_msg_id	4 (TARGET_UPDATE)
5	01	target_index	1

6-7	2D 00	bearing	45° (little-endian)
8-9	5E 01	PROXIMITY	350 (little-endian, i16; in proximity_unit , i.e. meters)
10	07	descriptor_len	7
11-17	57 41 59 50 54 2D 31	descriptor	"WAYPT-1"

► Step 5 — TARGET_UPDATE for target 2: WAYPT-2 at 270°, 120 m

Inner payload 14 bytes (LN = 0E).

```
00 03 01 0E 04 02 0E 01 78 00 07 57 41 59 50 54 2D 32
```

Byte(s)	Hex	Field	Value
4	04	class_msg_id	4 (TARGET_UPDATE)
5	02	target_index	2
6-7	0E 01	bearing	270° (little-endian)
8-9	78 00	PROXIMITY	120 (little-endian, i16; in proximity_unit , i.e. meters)
10	07	descriptor_len	7
11-17	57 41 59 50 54 2D 32	descriptor	"WAYPT-2"

► Step 6 — Clear target 1

The Device clears target index 1 by sending bearing = 0xFFFF ; the optional fields and descriptor are omitted. Inner payload 4 bytes (LN = 04).

```
00 03 01 04 04 01 FF FF
```

Byte(s)	Hex	Field	Value
4	04	class_msg_id	4 (TARGET_UPDATE)
5	01	target_index	1
6-7	FF FF	bearing	0xFFFF (clear)

The Host removes the arrow for target 1 from the display. Target 2 continues to be streamed.